National University of Computer and Emerging Sciences

AL-2002: Artificial Intelligence

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Objectives

After performing this lab, students shall be able to understand implementation of the following :

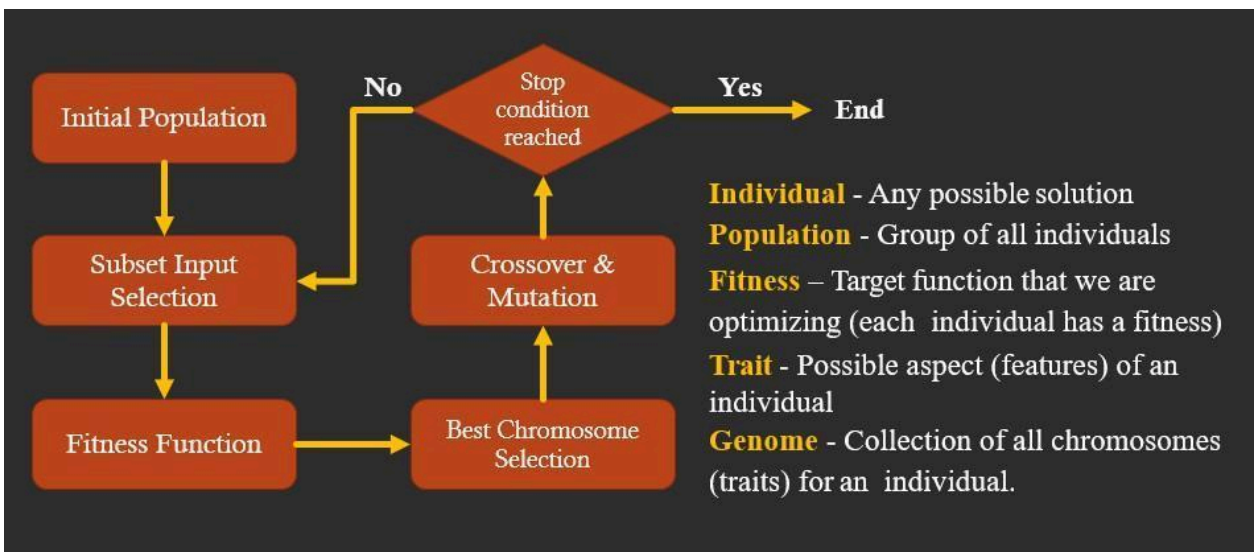
- ✔ Genetic Algorithm

1 Genetic Algorithm

A genetic algorithm (or GA) is a search technique used in computing to find true or approximate solutions to optimization and search problems. (GA)s are categorized as global search heuristics. (GA)s are a particular class of evolutionary algorithms that use techniques inspired by evolutionary biology such as inheritance, mutation, selection, and crossover (also called recombination).

- The evolution usually starts from a population of randomly generated individuals and happens in generations.
- In each generation, the fitness of every individual in the population is evaluated, multiple individuals are selected from the current population (based on their fitness), and modified to form a new population.
- The new population is used in the next iteration of the algorithm.
- The algorithm terminates when either a maximum number of generations has been produced, or a satisfactory fitness level has been reached for the population.

1.1 Genetic Algorithm Flow



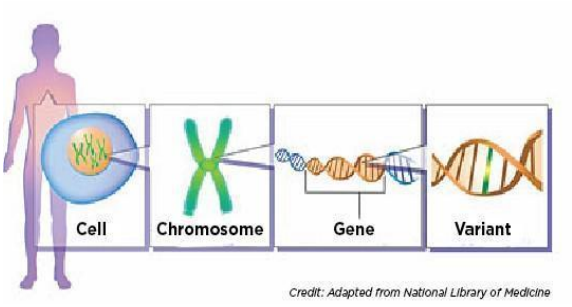
1.2 Population

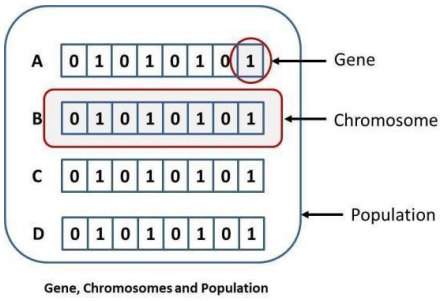
Biology

- Collection of individuals.

Algorithm

- Collection of states.





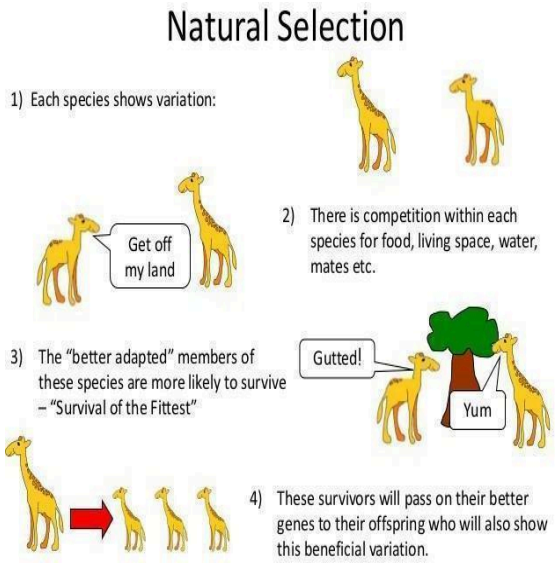
1.3 Fitness and Selection

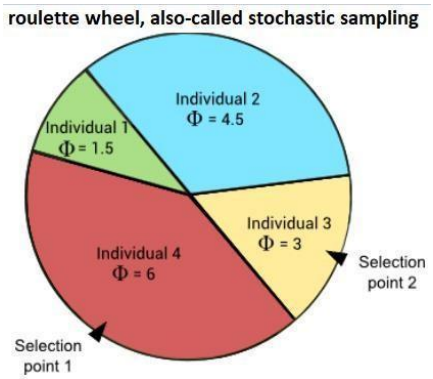
Biology

- More healthy, less prone to diseases.
- Selecting species that are the most biologically fit.

Algorithm

- Closest to the final solution.
- Selecting states that are closest to the solution (Fittest).



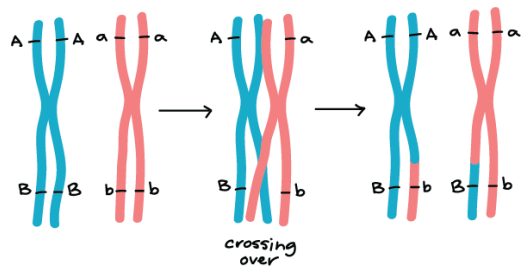


	Selection error	Rank	Fitness
Individual 1	0.9	1	1.5
Individual 2	0.6	3	4.5
Individual 3	0.7	2	3.0
Individual 4	0.5	4	6.0

1.4 Mating / Crossover / Generating Generations

Biology

- Mating or Reproducing



Algorithm

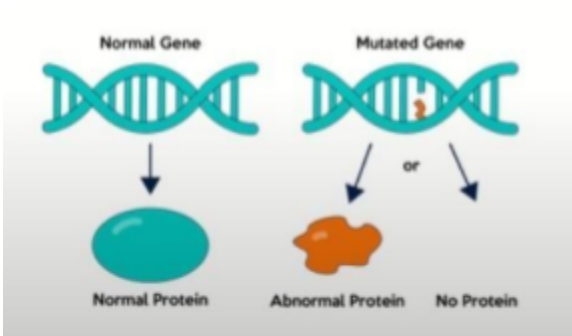
- Interchanging values between selected states.

Individual 3	1	1	0	0	0	1
Individual 4	0	1	0	1	0	0
Offspring 1	0	1	0	1	0	1
Offspring 2	1	1	0	1	0	1
Offspring 3	0	1	0	1	0	1
Offspring 4	1	1	0	0	0	0

1.5 Mutation

Biology

- Change or variation.



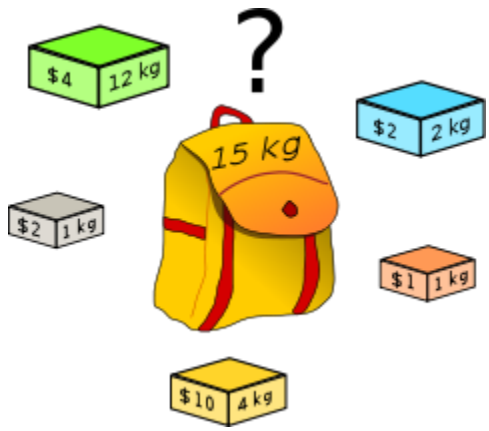
Algorithm

- Alteration.

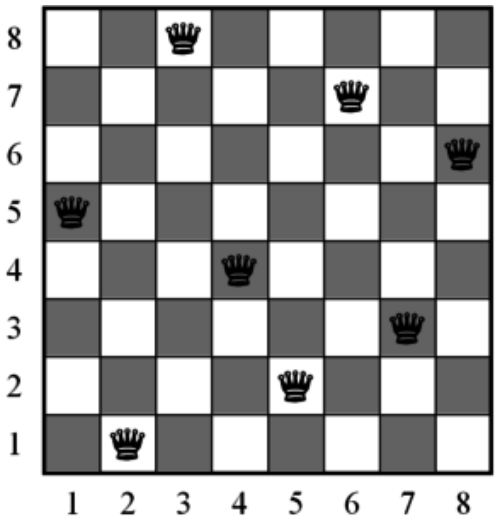
2 Problem Solving using Genetic Algorithm

Famous problems which use Genetic Algorithms:

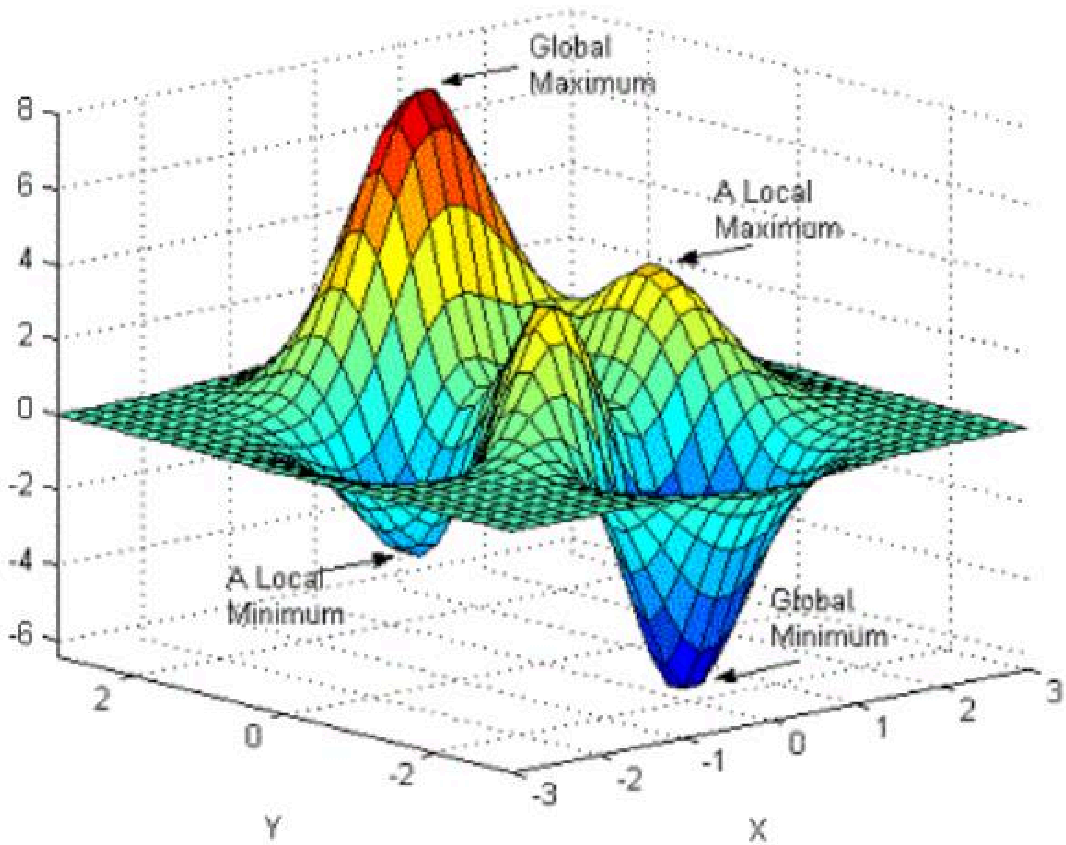
<u>KnapSack Problem</u>



<u>8 Queen Problem</u>



Optimization Problems/ Finding Global Maxima of a Function



Let's try to find optimized solution of a function:
Execute the attached notebook for this problem and follow the step by step details below for understanding.
Let's have random expression which evaluates to 25:
$6x^3 + 9y^2 + 90z^1 = 25$

Problem: What could be the best possible values of x, y and z that could satisfy the above expression.
For this purpose, we make function in python such that:

```
def evaluateExpression(x, y, z):
  return 6 * x ** 3 + 9 * y ** 2 + 90 * z - 25
```

This function that evaluates the expression to 0 if the answer of expression " 6 * x ** 3 + 9 * y ** 2 + 90 * z " is 25. So that means we need the most suitable values of x, y and z so that we could achieve our target value which is 25 in this case.

Now we need to apply Genetic Algorithm concepts to solve this problem.

Step1: Population of Solutions.
Population is generated entirely from random numbers let say up to 1000 individuals.

```
# generate solutions population
import random

solutions = []
for counter in range(1000):
  solutions.append((random.uniform(0, 1000), random.uniform(0, 1000), rand
om.uniform(0, 1000)))
```

Step2: Fitness function

So the fittest solution will be the one, which evaluates the expression to "0". Otherwise, the best solution will be closest to zero. Therefore, the fitness in this case is the highest if the solution is closest to zero. Hence, we can return the highest fitness value to those solutions, which are closest to 0.

```
def fitness(x, y, z):
  ans = evaluateExpression(x, y, z)

  if ans == 0:
    return 99999
  else:
    return abs(1 / ans)
```

Step3: Mating, Crossover or Generating the Generations
During each generations, further sub steps performed such as:
Step 3.1: Selection of top ranked solutions
Step 3.2: Mutation or slight changes or variation in values of solution.
P.S. Here for the sake of analogy if solution can be considered as chromosome then variable values can be considered as genes)

```
for generation_count in range(10000):
  rankedSolutions = []
  # fitness step
  for solution in solutions:
    rankedSolutions.append((fitness(solution[0], solution[1], solution[2])
, solution))
  rankedSolutions.sort()
  rankedSolutions.reverse()
  print(f"=== Generation {generation_count} best solutions ====")
  print(rankedSolutions[0])

  if rankedSolutions[0][0] > 999:
    break

  bestSolution = rankedSolutions[:100]
  # print(bestSolution)

  # selection step
  variables = []
  for solution in bestSolution:
    variables.append(solution[1][0]) # variable x
```

```
    variables.append(solution[1][1]) # variable y
    variables.append(solution[1][2]) # variable z

  newGeneration = []
  # mutation step
  for counter in range(1000):
    x = random.choice(variables) * random.uniform(0.99, 1.01)
    y = random.choice(variables) * random.uniform(0.99, 1.01)
    z = random.choice(variables) * random.uniform(0.99, 1.01)

    newGeneration.append((x, y, z))

  solutions = newGeneration
```

Note: After running this code given in Jupyter Notebook, confirm the values from best optimized solution into given expression to verify.